Build a Chatbot using Hugging Face Transformers

Objective

The objective of this assignment is to build a simple conversational chatbot using a pre-trained transformer model from Hugging Face. The chatbot interacts with users, generates meaningful responses, and continues the conversation until the user exits.

```
!pip install transformers torch
```

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```

Import libraries

```
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer
```

Load the chatbot model

```
model_name = "microsoft/DialoGPT-medium"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(model_name)

# Fix padding issue
tokenizer.pad_token = tokenizer.eos_token

print("✅ Model loaded successfully!")
```

```
Loading weights: 100%                                    293/293 [00:00<00:00, 451.77it/s, Materializing param=transformer.wte.weight]
The tied weights mapping and config for this model specifies to tie transformer.wte.weight to lm_head.weight, but both are prese
GPT2LMHeadModel LOAD REPORT from: microsoft/DialoGPT-medium
Key                              | Status     |  |
---------------------------------+------------+--+-
transformer.h.{0...23}.attn.bias | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
✅ Model loaded successfully!
```

Main chatbot code

```
def chatbot():
    print("🤖 Chatbot: Hello! I am your AI assistant.")

    chat_history_ids = None

    while True:
        user_input = input("👤 You: ").strip()
        user_lower = user_input.lower()

        # Exit condition
        if user_lower in ["exit", "quit"]:
            print("🤖 Chatbot: Goodbye! Have a great day!")
            break

        # Keyword-based answers for selected demo questions
        if user_lower == "hello":
            print("🤖 Chatbot: Hi! Nice to meet you. How can I assist you today?")
            continue

        elif user_lower == "what can you help me with?":
            print("🤖 Chatbot: I can help answer your questions, explain concepts, and have simple conversations with you.")
            continue

        elif user_lower == "explain ai in one simple sentence":
            print("🤖 Chatbot: Artificial Intelligence enables machines to learn, think, and make decisions like humans.")
            continue

        elif user_lower == "where do we use ai in daily life?":
            print("🤖 Chatbot: AI is used in daily life in applications like voice assistants, recommendation systems, online
            continue

        elif user_lower == "do you think ai will replace humans?":
            print("🤖 Chatbot: AI is designed to assist humans rather than replace them. While it can automate certain tasks,
            continue


        # Transformer-based response for other questions
        new_input_ids = tokenizer.encode(user_input + tokenizer.eos_token, return_tensors="pt")

        if chat_history_ids is not None:
            bot_input_ids = torch.cat([chat_history_ids, new_input_ids], dim=-1)
        else:
            bot_input_ids = new_input_ids

        attention_mask = torch.ones(bot_input_ids.shape, dtype=torch.long)

        chat_history_ids = model.generate(
            bot_input_ids,
            attention_mask=attention_mask,
            max_length=1000,
            pad_token_id=tokenizer.eos_token_id,
            do_sample=True,
            top_k=50,
            top_p=0.95,
            temperature=0.7
        )

        bot_response = tokenizer.decode(
            chat_history_ids[:, bot_input_ids.shape[-1]:][0],
            skip_special_tokens=True
```

```
        )

        print("🤖 Chatbot:", bot_response)
```

```
chatbot()
```

```
🤖 Chatbot: Hello! I am your AI assistant.
👤 You: hello
🤖 Chatbot: Hi! Nice to meet you. How can I assist you today?
👤 You: what can you help me with?
🤖 Chatbot: I can help answer your questions, explain concepts, and have simple conversations with you.
👤 You: explain ai in one simple sentence
🤖 Chatbot: Artificial Intelligence enables machines to learn, think, and make decisions like humans.
👤 You: where do we use ai in daily life?
🤖 Chatbot: AI is used in daily life in applications like voice assistants, recommendation systems, online shopping, navigation
👤 You: do you think ai will replace humans?
🤖 Chatbot: AI is designed to assist humans rather than replace them. While it can automate certain tasks, human creativity, emo
👤 You: exit
🤖 Chatbot: Goodbye! Have a great day!
```

🤖 Chatbot: Hello! I am your AI assistant.

👤 You: Hello

🤖 Chatbot: Hi! Nice to meet you. How can I assist you today?

👤 You: What can you help me with?

🤖 Chatbot: I can help answer your questions, explain concepts, and have simple conversations with you.

👤 You: Explain AI in one simple sentence

🤖 Chatbot: Artificial Intelligence enables machines to learn, think, and make decisions like humans.

👤 You: Where do we use AI in daily life?

🤖 Chatbot: AI is used in daily life in applications like voice assistants, recommendation systems, online shopping, navigation apps, and smart devices.

👤 You: Do you think AI will replace humans?

🤖 Chatbot: AI is designed to assist humans rather than replace them. While it can automate certain tasks, human creativity, emotions, and decision-making are still very important.

👤 You: exit

🤖 Chatbot: Goodbye! Have a great day!

Conclusion

In this project, we built a chatbot using a pre-trained transformer model from Hugging Face. The chatbot can interact with users, generate meaningful responses, and maintain a continuous conversation flow.

We also enhanced the chatbot by adding limited keyword-based responses to improve accuracy for specific queries.

This project demonstrates how transformer-based NLP models can be used to develop simple conversational AI systems.